

MAT 320 HW07

Aditya Pahuja

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1 Page of Lemmas

The following two facts appeared frequently enough within the problems and have long enough proofs that I am moving them into this section for more convenient referencing.

Lemma L1 (Open balls are open). Let (X, d) be a metric space. For any $\varepsilon > 0$, the open ball $B_X(x, \varepsilon)$ is actually an open set in X (thus open balls are worthy of their name).

Proof. For any $y \in B_X(x, \varepsilon)$, consider the open ball

$$B_X(y, \varepsilon - d_X(x, y))$$

Any $z \in B_X(y, \varepsilon - d_X(x, y))$ satisfies

$$d_X(x, z) \leq d_X(x, y) + d_X(y, z) < d_X(x, y) + \varepsilon - d_X(x, y) = \varepsilon.$$

Thus, every element of $B_X(y, \varepsilon - d_X(x, y))$ is in $B_X(x, \varepsilon)$, meaning that, for every $y \in B_X(x, \varepsilon)$, we have found an open ball $B_X(y, \varepsilon - d_X(x, y))$ centered at y that is contained inside $B_X(x, \varepsilon)$, proving that $B_X(x, \varepsilon)$ is indeed open in X . $\square$

Lemma L2 (Linear combinations are continuous). Let $\mathbf{x}$ and $\mathbf{y}$ be arbitrary elements of $\mathbb{R}^n$. Then, the function $\gamma: [0, 1] \rightarrow \mathbb{R}^n$ given by

$$\gamma(t) = (1 - t)\mathbf{x} + t\mathbf{y}$$

is continuous (where $\mathbb{R}^n$ has the standard metric, which we will call d).

Proof. Let t be in $[0, 1]$; we need to show that γ is continuous at t . Recall also that $\|\mathbf{x}\| = d(\mathbf{x}, \mathbf{0})$ is the norm of $\mathbf{x}$, so that $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, and that $\|c\mathbf{x}\| = |c| \cdot \|\mathbf{x}\|$ for any real c .

Given $\varepsilon > 0$, let $\delta = \frac{\varepsilon}{d(\mathbf{x}, \mathbf{y})} > 0$. If $|s - t| < \delta$, then we have

$$\begin{aligned} d(\gamma(s), \gamma(t)) &= \|((1 - s)\mathbf{x} + s\mathbf{y}) - ((1 - t)\mathbf{x} + t\mathbf{y})\| \\ &= \|(s - t)\mathbf{y} + (t - s)\mathbf{x}\| \\ &= |s - t| \cdot \|\mathbf{y} - \mathbf{x}\| \\ &= |s - t| \cdot d(\mathbf{x}, \mathbf{y}) \\ &< \frac{\varepsilon}{d(\mathbf{x}, \mathbf{y})} \cdot d(\mathbf{x}, \mathbf{y}) = \varepsilon. \end{aligned}$$

That is, we just found a $\delta > 0$ such that $|s - t| < \delta$ implies $d(\gamma(s), \gamma(t)) < \varepsilon$, hence γ is continuous at all $t \in [0, 1]$. $\square$

Lemma L3 (Path connectedness is an equivalence relation). Given a metric space (X, d) , we say that $p \sim q$ if p and q are connected by a path, i.e. there exists a continuous function $\gamma: [0, 1] \rightarrow X$ such that $\gamma(0) = p$ and $\gamma(1) = q$. Then, $\sim$ is an equivalence relation on X .

Proof. First, the relation is reflexive. To construct a path from a to itself, we can take $\gamma(t) = a$ for all $t \in [0, 1]$, which is obviously continuous.

Second, the relation is symmetric. If $a \sim b$, then there exists a path γ going from a to b . Noting that the map $t \mapsto 1 - t$ is continuous, we have

$$\hat{\gamma}(t) = \gamma(1 - t)$$

is a continuous function $\hat{\gamma}: [0, 1] \rightarrow X$ because it is a composition of continuous functions. It also satisfies $\hat{\gamma}(0) = \gamma(1) = b$ and $\hat{\gamma}(1) = \gamma(0) = a$, so $b \sim a$.

Third, the relation is transitive. If $a \sim b$ and $b \sim c$, then there exist paths γ_1 and γ_2 from a to b and from b to c respectively. To get a path from a to c , we can “glue” the paths together: define the function $\gamma: [0, 1] \rightarrow X$ given by

$$\gamma(t) = \begin{cases} \gamma_1(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ \gamma_2(2t - 1) & \text{if } \frac{1}{2} < t \leq 1. \end{cases}$$

Clearly γ satisfies $\gamma(0) = \gamma_1(0) = a$ and $\gamma(1) = \gamma_2(1) = c$. To show that γ is continuous we consider three cases.

- Let $0 < t < \frac{1}{2}$ and $\varepsilon > 0$. Knowing that γ_1 is continuous at $2t$, there exists $0 < \delta < 1 - 2t$ such that

$$|2x - 2t| < \delta \implies d(\gamma_1(2x), \gamma_1(2t)),$$

where $\delta < 1 - 2t$ guarantees that $|2x - 2t| < \delta$ implies $2x < 1$. This means that $\gamma(t)$ is continuous at t , since now we have

$$|x - t| < \frac{\delta}{2} \implies |2x - 2t| < \delta \implies d(\gamma_1(2x), \gamma_1(2t)) = d(\gamma(x), \gamma(t)) < \varepsilon,$$

where we can write $\gamma(x)$ in place of $\gamma_1(2x)$ because $2x < 1 \iff x < \frac{1}{2}$ is forced.

- Similarly, if $\frac{1}{2} < t \leq 1$, then, given $\varepsilon > 0$, we pick $0 < \delta < 2t - 1$ such that

$$|(2x - 1) - (2t - 1)| < \delta \implies d(\gamma_2(2x - 1), \gamma_2(2t - 1)) < \varepsilon$$

by continuity of γ_2 . Here, $\delta < 2t - 1$ guarantees that $|(2x - 1) - (2t - 1)| < \delta$ implies that $2x - 1 > 0$. This translates to the continuity of γ at t because it says that

$$\begin{aligned} |x - t| < \frac{\delta}{2} &\implies |(2x - 1) - (2t - 1)| < \delta \\ &\implies d(\gamma_2(2x - 1), \gamma_2(2t - 1)) = d(\gamma(x), \gamma(t)) < \varepsilon, \end{aligned}$$

where we can safely substitute $\gamma_2(2x - 1) = \gamma(x)$ because $2x - 1 > 0 \iff x > \frac{1}{2}$.

- Finally, we need to check that γ is continuous at $\frac{1}{2}$. We know that γ_1 is continuous at 1 and γ_2 is continuous at 0. Thus, we can say that, given $\varepsilon > 0$, there exists $\delta_1 > 0$ such that

$$|2x - 1| < \delta_1 \implies d(\gamma_1(2x), b) < \varepsilon$$

and there exists $\delta_2 > 0$ such that

$$|(2x - 1) - 0| < \delta_2 \implies d(\gamma_2(2x - 1), b) < \varepsilon,$$

noting that $\gamma(\frac{1}{2}) = \gamma_1(1) = \gamma_2(0) = b$. Therefore, if $\delta = \min(\delta_1, \delta_2)$, then

$$\left| x - \frac{1}{2} \right| < \frac{\delta}{2} \implies |2x - 1| < \delta \implies d(\gamma(x), b) < \varepsilon$$

where the second implication comes from the above two inequalities saying that if $x \leq \frac{1}{2}$ then $d(\gamma_1(2x), b) = d(\gamma(x), b) < \varepsilon$ and if $x > \frac{1}{2}$ then $d(\gamma_2(2x - 1), b) = d(\gamma(x), b) < \varepsilon$. This means that γ is continuous at $\frac{1}{2}$, as desired.

Thus, a and c are connected by a path, which means $a \sim c$. □

2 Solutions to problems

Problem 1. Let (X, d) be a metric space. Given $E \subseteq X$ define the **closure** of E as

$$\overline{E} = \{x \in X : x \text{ is a limit point of } E\} \subseteq X.$$

Explain why $E \subseteq \overline{E}$, and show that $\overline{E}$ equals the intersection of all closed subsets $F \subseteq X$ that contain E .

Solution. First, to show that $E \subseteq \overline{E}$, we need to show that every point of E is a limit point of E . This is obvious, though: for a given $x \in E$, we can find a sequence converging to x by taking the constant sequence $(x, x, x, \dots)$.

Next, let E' be the intersection of all closed subsets $F \subseteq X$ containing E ; we need to show that $E' = \overline{E}$. Note that E' is closed because it is the intersection of a collection of closed sets. Since E' contains E , E' must also contain every limit point of E because a set is closed if and only if it contains all of its limit points, which lets us conclude that $\overline{E} \subseteq E'$.

On the other hand, we wish to show that $E' \subseteq \overline{E}$ as well.

Claim 1.1 — $\overline{E}$ is a closed subset of X .

Proof. Let (x_n) be a sequence of points in $\overline{E}$ that converge to some $\ell \in X$, i.e. ℓ is a limit point of $\overline{E}$. For each $n \in \mathbb{N}$, since x_n is a limit point of E , we can find a sequence of points in E that converge to x_n ; in particular, we can find a point $y_n \in E$ such that $d(x_n, y_n) < \frac{1}{n}$. By the triangle inequality,

$$d(\ell, y_n) \leq d(\ell, x_n) + d(x_n, y_n) < d(\ell, x_n) + \frac{1}{n}$$

Let $\varepsilon > 0$ be an arbitrary positive real and let N be a positive integer greater than $\frac{1}{\varepsilon}$. Then, $\varepsilon - \frac{1}{N}$ is a positive real number, hence $d(\ell, x_n) < \varepsilon - \frac{1}{N}$ is eventually true, say when $n > M$. This means

$$d(\ell, x_n) + \frac{1}{n} < \varepsilon - \frac{1}{N} + \frac{1}{n}$$

is true when $n > M$, and the expression on the right-hand side is smaller than ε when $n > N$, so the inequality

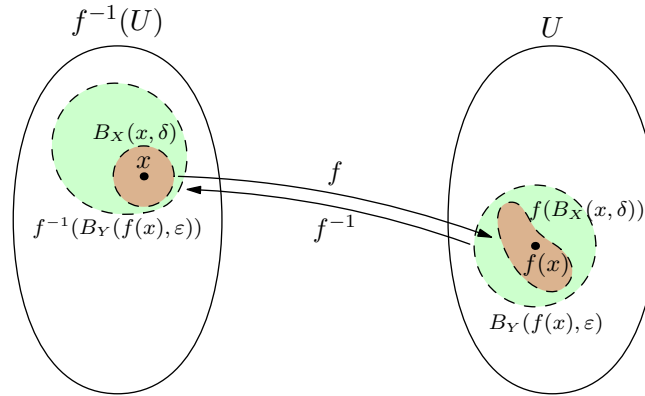
$$d(\ell, y_n) < d(\ell, x_n) + \frac{1}{n} < \varepsilon - \frac{1}{N} + \frac{1}{n} < \varepsilon$$

holds eventually, namely when $n > \max(M, N)$. In particular, (y_n) converges to ℓ , so ℓ is a limit point of E (because $y_n \in E$ for all n). This means $\overline{E}$ contains all of its limit points, so it is closed. $\square$

This means $\overline{E} \supset E$ is a closed set containing E , which means E' must be a subset of $\overline{E}$ because E' is a subset of every closed set containing E by definition. This lets us conclude that $E' = \overline{E}$ as desired. $\square$

Problem 2 (Writing exercise). Show that a function between metric spaces $X \rightarrow Y$ is continuous if and only if $f^{-1}(U)$ is *open* for each *open* set $U \subseteq Y$. Show that the same is true if in this statement you replace the word *open* with the word *closed*.

Solution. Here's a picture:



First, suppose f is continuous. For an open $U \subseteq Y$, let $x \in f^{-1}(U)$ and let $B_Y(f(x), \varepsilon)$ be contained in U . Then, there exists a $\delta > 0$ so that

$$f(B_X(x, \delta)) \subseteq B_Y(f(x), \varepsilon)$$

since being continuous at x implies the existence of $\delta > 0$ such that $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$. This means that $B_X(x, \delta) \subseteq f^{-1}(U)$, i.e. for each $x \in f^{-1}(U)$, we can find an open ball centered at x contained in U , so $f^{-1}(U)$ is open.

Conversely, suppose $f^{-1}(U)$ is open in X for all open $U \subseteq Y$. By Lemma L1, the open ball $B_Y(f(x), \varepsilon)$ is open in Y . Thus, $f^{-1}(B_Y(f(x), \varepsilon))$ is open in X and contains x . This means we can find an open ball $B_X(x, \delta) \subseteq f^{-1}(B_Y(f(x), \varepsilon))$ due to openness, so, for each $\varepsilon > 0$, there exists $\delta > 0$ such that

$$f(B_X(x, \delta)) \subseteq B_Y(f(x), \varepsilon),$$

so f is continuous.

Now, we prove the second part (replacing “open” with “closed”). The statement

$$E \subseteq Y \text{ is closed} \implies f^{-1}(E) \subseteq X \text{ is closed}$$

is equivalent to

$$Y \setminus E \subseteq Y \text{ is open} \implies f^{-1}(Y \setminus E) = X \setminus f^{-1}(E) \subseteq X \text{ is open.}$$

Then, $Y \setminus E$ is open iff $E = Y \setminus U$ for an open $U \subseteq Y$, so the line above is equivalent to

$$U \subseteq Y \text{ is open} \implies f^{-1}(U) \subseteq X \text{ is open.}$$

This is then equivalent to f being continuous by the work done above, so we are done. $\square$

Problem 3. Show that the composition of two continuous functions $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ between metric spaces gives rise to a continuous function $g \circ f: X \rightarrow Z$.

Solution. Let V be an open subset of Z . Then, letting $U = g^{-1}(V)$,

$$(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V)) = f^{-1}(U) \subseteq X.$$

By Problem 2, since g and f are continuous, $U = g^{-1}(V)$ is open in Y and $f^{-1}(U)$ is open in X , so, by Problem 2 again, $f \circ g$ is continuous. $\square$

Problem 4. Let (X, d_X) be a metric space. Show that a subset $A \subseteq E$ is open in the induced metric d_E if and only if there exists an open set $U \subseteq X$ such that $A = U \cap E$. Show that if E is open in X then the open subsets (E, d_E) are exactly the open subsets of X that are contained in E . Show that there exist open subsets of $[0, +\infty)$ that are not open in $\mathbb{R}$.

Solution. We first show that $A \subseteq E$ is open if and only if $A = U \cap E$ where U is open in X .

If $A \subseteq E$ is open with respect to the induced metric, then, for each $x \in A$, there exists an $\varepsilon_x > 0$ such that the open ball

$$B_E(x, \varepsilon_x) = \{y \in E : d_E(x, y) < \varepsilon_x\}$$

is a subset of A . We can therefore write

$$A = \bigcup_{x \in A} B_E(x, \varepsilon_x),$$

as, on one hand, each of the balls is contained in A , thus so is their union, so the right-hand side is a subset of A ; and, on the other hand, each $x \in A$ is in the open ball $B_E(x, \varepsilon_x)$, so the union contains A .

However, $B_E(x, \varepsilon_x)$ can be written as $B_X(x, \varepsilon_x) \cap E$, noting that $d_E(x, y) = d_X(x, y)$ for $x, y \in E$. We can thus write the above as

$$A = \bigcup_{x \in A} (B_X(x, \varepsilon_x) \cap E) = E \cap \left(\bigcup_{x \in A} B_X(x, \varepsilon_x) \right).$$

Thus, since each open ball is open in X by Lemma L1, the union of the open balls is an open set U , and so we have found an open set U such that $A = E \cap U$.

Conversely, if $U \subseteq X$ is open, then for each $x \in U$, there exists $\varepsilon_x > 0$ such that the ball $B_X(x, \varepsilon_x)$ is a subset of U . This means that, if $x \in U \cap E$,

$$B_E(x, \varepsilon_x) = B_X(x, \varepsilon_x) \cap E \subseteq U \cap E.$$

That is, for each x in the intersection of U and E , we can find an open ball in the induced metric d_E centered at x that is contained in $U \cap E$, which is what it means for $U \cap E$ to be open in E .

If E is open in X , then $A \subseteq E$ is open in E if and only if $A = E \cap U$ for some U that is open in X . However, this means that A , being the intersection of two open subsets of X , is also open in X , which means A is open in E if and only if A is open in X .

Finally, the set $U = [0, 1)$ is not open in $\mathbb{R}$ because any open ball centered at 0 must contain numbers outside U , but it is an open subset of $[0, +\infty)$ because U is the intersection of the open set $(-1, 1) \subseteq \mathbb{R}$ with $[0, +\infty)$. $\square$

Problem 5. Explain the following statement that was made during class: $E \subseteq X$ is disconnected if and only if there are open sets $A, B \subseteq X$ such that $E \subseteq A \cup B$, $E \cap A \neq \emptyset$, $E \cap B \neq \emptyset$, and $E \cap (A \cap B) = \emptyset$. How is this statement simplified if E is open?

Solution. By definition, $E \subseteq X$ is disconnected if and only if there exist nonempty disjoint open subsets A' and B' of E whose union is E . By Problem 5, we know that $A' = A \cap E$ and $B' = B \cap E$ where A and B are open in X . We can now verify the criteria for E being disconnected:

- The condition that A' and B' are both nonempty translates to $A \cap E$ and $B \cap E$ both being nonempty.
- The condition that A' and B' are disjoint translates to $(A \cap E) \cap (B \cap E) = (A \cap B) \cap E$ being the empty set.
- Finally, $A' \cap B' = E$ translates to $(A \cap E) \cup (B \cap E) = (A \cup B) \cap E = E$, which is true if and only if $E \subseteq A \cup B$.

If E is open, then Problem 5 says that A' and B' will also be open in X , so the condition for disconnectedness is just that there exist disjoint open subsets of X whose union is E . $\square$

Problem 6. Let (X, d) be a metric space. Show that if $D \subseteq X$ is connected then its closure is also connected. Use this to show that, given (a, b) is connected, $(a, b]$, $[a, b)$, and $[a, b]$ are also connected.

Solution. We prove the contrapositive statement that $\overline{D}$ being disconnected implies D is disconnected. If $\overline{D}$ is disconnected, then, by Problem 5, there exist open subsets of X , say A and B , whose union contains $\overline{D}$ such that $A \cap \overline{D}$ and $B \cap \overline{D}$ are both nonempty but $A \cap B \cap \overline{D}$ is empty. Since $D \subseteq \overline{D}$ by Problem 1, we already know that

$$A \cap B \cap D = \emptyset, \quad D \subseteq A \cup B.$$

To show that D is disconnected we need to prove that $A \cap D$ and $B \cap D$ are nonempty. We will prove that $A \cap D$ is nonempty; the proof for $B \cap D$ is identical.

Suppose that x is an element of $A \cap \overline{D}$. Since $A \cap \overline{D}$ is open in $\overline{D}$, there exists an $\varepsilon > 0$ such that the open ball $B_{\overline{D}}(x, \varepsilon)$ is a subset of $A \cap \overline{D}$. Then, x is a limit point of D , so we can find a sequence of points in D converging to x . In particular, we can find a $y \in D$ such that $y \in B_{\overline{D}}(x, \varepsilon)$, which means that y is in $A \cap \overline{D}$. This means that $y \in A$, so $y \in A \cap D$, and thus $A \cap D$ is nonempty, as desired.

Now, we are ready to show that all kinds of intervals are connected! First, for closed intervals, we show that $[a, b]$ is the closure of (a, b) over $\mathbb{R}$. Indeed, a and b are limit points of (a, b) because the sequences

$$a_n = a + \frac{b-a}{n}, \quad b_n = b - \frac{b-a}{n}$$

for $n > 1$ converge to a and b respectively and consist only of points in (a, b) (since $a < a_n < a + b - a = b$ and $a = b - (b - a) < b_n < b$), and obviously all points of (a, b) are in its closure by Problem 1. Moreover, (a, b) has no other limit points because, if a sequence (x_n) satisfies $a < x_n < b$ for all n , then taking limits as n goes to infinity shows that

$$a \leq \lim_{n \rightarrow \infty} x_n \leq b$$

if the limit exists. Thus $[a, b] = \overline{(a, b)}$ is connected.

For half-open intervals, we use a little trick.

Lemma 6.1. Consider an open set $U \subseteq X$. A subset $E \subseteq U$ is connected in the metric space (U, d_U) if and only if it is connected in (X, d) .

Proof. First, suppose E is disconnected in (X, d) . Then, there exist open subsets of X , say A and B , such that

$$E \subseteq A \cup B, \quad E \cap A \neq \emptyset, \quad E \cap B \neq \emptyset, \quad E \cap (A \cap B) = \emptyset.$$

Then, $A' = A \cap U$ and $B' = B \cap U$ are open in U , and, since $E \subseteq U$, we can say that

- E is a subset of $A' \cup B' = (A \cup B) \cap U$,
- $E \cap A' = E \cap A \neq \emptyset$,
- $E \cap B' = E \cap B \neq \emptyset$, and
- $E \cap (A' \cap B') \subseteq E \cap (A \cap B) = \emptyset$ (which obviously means $E \cap (A' \cap B') = \emptyset$).

Thus, E is also disconnected in (U, d_U) .

For the other direction, suppose E is disconnected in (U, d_U) . Then, E is a union of disjoint sets A and B that are open in U . Problem 4 shows us that, since U is open, A and B are open in X , so we just treat A and B as open subsets of X instead of U to say that E is connected in X . $\square$

Now consider the subspaces of $\mathbb{R}$ given by $A = (a, +\infty) \subseteq \mathbb{R}$ and $B = (-\infty, b) \subseteq \mathbb{R}$, which are open intervals and thus open. Clearly, (a, b) is still connected within each of these spaces. The closure of (a, b) within A is now $[a, b] \cap A = (a, b]$, since $a \notin A$ (but all the other limit points are exactly preserved), and similarly the closure of (a, b) within B is $[a, b)$. The lemma tells us that the connectedness of these half-open intervals within the respective subspace is equivalent to their connectedness in $\mathbb{R}$ because A and B are open, so $(a, b]$ and $[a, b)$ are connected in $\mathbb{R}$, too! $\square$

Problem 7. Let $f: X \rightarrow Y$ be a continuous function between metric spaces. Show that if $E \subseteq X$ is path connected, then $f(E) \subseteq Y$ is also path connected.

Proof. Given two points a and b in $f(E)$, there exist a' and b' in E such that $f(a') = a$ and $f(b') = b$. Let $\gamma: [0, 1] \rightarrow E$ be a path between a' and b' , so $\gamma(0) = a'$, $\gamma(1) = b'$, and γ is continuous. Then, define $\Gamma: [0, 1] \rightarrow f(E)$ by $\Gamma(t) = f(\gamma(t))$. This is a composition of continuous functions, so it is continuous by Problem 3, and

$$\Gamma(0) = f(\gamma(0)) = f(a') = a, \quad \Gamma(1) = f(\gamma(1)) = f(b') = b.$$

Thus, we have constructed a path Γ between a and b , so any pair of points in $f(E)$ has a path between them, and thus $f(E)$ is path connected. $\square$

Problem 8. A subset $C \subseteq \mathbb{R}^n$ is called **convex** if the line segment connecting any two points of C lies completely within C . Show that any convex subset of $\mathbb{R}^n$ is connected. Is it true that a subset of $\mathbb{R}^n$ is connected if and only if it is convex?

Solution. Let $C \subseteq \mathbb{R}^n$ be convex and let $\mathbf{x}, \mathbf{y} \in C$. The line segment between $\mathbf{x}$ and $\mathbf{y}$ is the set

$$\ell(\mathbf{x}, \mathbf{y}) = \{(1-t)\mathbf{x} + t\mathbf{y} : t \in [0, 1]\} \subseteq C.$$

Define $\gamma: [0, 1] \rightarrow C$ by

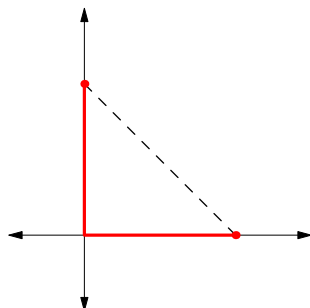
$$\gamma(t) = (1-t)\mathbf{x} + t\mathbf{y}.$$

This is a continuous function by Lemma L2 such that $\gamma(0) = \mathbf{x}$ and $\gamma(1) = \mathbf{y}$, so it is a path between $\mathbf{x}$ and $\mathbf{y}$. Thus, we can find a path between any two points of C , so C is path connected, meaning it is connected.

The converse of this statement isn't true — there exist connected sets in $\mathbb{R}^n$ that aren't convex. For example, in $\mathbb{R}^2$, we will show that

$$S = \{(a, 0) : a \in [0, 1]\} \cup \{(0, a) : a \in [0, 1]\}$$

is connected but not convex.



Clearly S isn't convex, because the segment between $(0, 1)$ and $(1, 0)$ contains points outside of S , like $(\frac{1}{2}, \frac{1}{2})$.

However, we will show that it is path connected and thus connected. Let $\mathbf{a}$ and $\mathbf{b}$ be two points in S .

- First, we will show that $\mathbf{0}$ is connected to every point $\mathbf{x} \in S$. Define $\gamma: [0, 1] \rightarrow S$ by $\gamma(t) = t\mathbf{x}$. If $\mathbf{x} \in S$ then $t\mathbf{x} \in S$ because $\mathbf{x} = (a, 0)$ or $(0, a)$ for some $a \in [0, 1]$, and $t\mathbf{x} = (ta, 0)$ or $(0, ta)$, in which case $ta \in [0, 1]$ because $0 \leq t \leq 1$. Additionally, writing $\gamma(t) = (1-t)\mathbf{0} + t\mathbf{x}$, we can use Lemma L2 to conclude that γ is continuous, and we can obviously see that $\gamma(0) = \mathbf{0}$ and $\gamma(1) = \mathbf{x}$. We have thus verified all the conditions for γ to be a path between the two points, so $\mathbf{0}$ and $\mathbf{x}$ are connected by a path.
- Now, $\mathbf{a}$ and $\mathbf{b}$ are connected by a path to $(0, 0)$, so Lemma L3 lets us say that, since path connectedness is transitive, $\mathbf{a}$ and $\mathbf{b}$ are also connected by some path.

Thus S is (path) connected, but not convex. □

Problem 9. Show that an open set $E \subseteq \mathbb{R}^n$ is connected if and only if it is path connected.

Solution. We have already shown in class that E being path connected implies that it is connected. We wish to prove the converse statement that if E isn't path connected, then it is not connected.

Claim 9.1 — Let $\varepsilon > 0$ and $\mathbf{x} \in \mathbb{R}^n$. If $\mathbf{y} \in B_{\mathbb{R}^n}(\mathbf{x}, \varepsilon)$, then there exists a path connecting $\mathbf{x}$ and $\mathbf{y}$.

Proof. Noting that we are working in $\mathbb{R}^n$, we consider the line segment connecting $\mathbf{x}$ and $\mathbf{y}$, consisting of the points $\gamma(t) = (1-t)\mathbf{x} + t\mathbf{y}$ with $0 \leq t \leq 1$. Observe that

$$\begin{aligned} d_{\mathbb{R}^n}(\mathbf{x}, \gamma(t)) &= d(\mathbf{x}, (1-t)\mathbf{x} + t\mathbf{y}) \\ &= \|(1-t)\mathbf{x} + t\mathbf{y} - \mathbf{x}\| = \|t(\mathbf{y} - \mathbf{x})\| \\ &= t \cdot d_{\mathbb{R}^n}(\mathbf{x}, \mathbf{y}) < t\varepsilon \leq \varepsilon, \end{aligned}$$

noting that $t\varepsilon \leq \varepsilon$ because $0 \leq t \leq 1$. Thus, the line segment between $\mathbf{x}$ and $\mathbf{y}$ is always contained in the open ball, so we can just take this segment as a path between the two points, noting that Lemma L2 tells us that $\gamma(t)$ is continuous. $\square$

As in Lemma L3, define the relation $\sim$ on E by saying that $a \sim b$ if and only if there is a path inside E that connects a to b . Let $\mathcal{U}$ be the collection of equivalence classes of E under $\sim$; the elements of $\mathcal{U}$ form a partition of E . We must have more than one element in $\mathcal{U}$ because if there is only one equivalence class under $\sim$, then any two elements of E are connected by a path, i.e. E is path connected, which contradicts our assumption.

Claim 9.2 — Each element of $\mathcal{U}$ is an open subset of E .

Proof. Let U be an element of $\mathcal{U}$ and let x be in U . We know that, because E is open in $\mathbb{R}^n$, there exists $\varepsilon > 0$ such that the open ball $B_{\mathbb{R}^n}(x, \varepsilon)$ is contained in E . By Claim 9.1, each element of this ball can be connected to x via a path, so each element of this ball is in U , meaning

$$B_{\mathbb{R}^n}(x, \varepsilon) \subseteq U.$$

This holds for every $x \in U$, so U is open in $\mathbb{R}^n$ and thus also open in E (because $U \subseteq E$ implies $U = E \cap U$), which is what we wanted to show. $\square$

To finish, $\mathcal{U}$ is a partition of E into (more than one) open sets — that is, E is the union of a collection of pairwise disjoint open sets. Let U_1 be an arbitrary element of $\mathcal{U}$ (which is obviously nonempty) and let U_2 be the union of all the other elements of $\mathcal{U}$, so

$$U_1 \cup U_2 = \bigcup_{U \in \mathcal{U}} U = E.$$

Because $\mathcal{U}$ has more than one element, U_2 is nonempty. Additionally, U_2 is a union of open sets and thus also open in E , and it is disjoint from U_1 because the elements of $\mathcal{U}$ are pairwise disjoint. Therefore, we have found sets U_1 and U_2 that are open in E , nonempty and disjoint, and whose union is E , which proves that E is disconnected. $\square$

Problem 10. Show that if X and Y are path connected metric spaces then so is $X \times Y$. Extra: do the same but replacing *path connected* with *connected*.

Solution. First, suppose X and Y are path connected, and let (a_1, b_1) and (a_2, b_2) be in $X \times Y$. We know that there exist paths $\gamma_X: [0, 1] \rightarrow X$ connecting a_1 and a_2 , and $\gamma_Y: [0, 1] \rightarrow Y$ connecting b_1 and b_2 . Then, I claim that $\gamma: [0, 1] \rightarrow X \times Y$, given by

$$\gamma(t) = (\gamma_X(t), \gamma_Y(t))$$

is a path between (a_1, b_1) and (a_2, b_2) . Obviously $\gamma(0) = (a_1, b_1)$ and $\gamma(1) = (a_2, b_2)$, so we only need to prove that γ is continuous with respect to the product metric $d_{X \times Y}$ (which we can recall is defined as $d_{X \times Y}((a_1, b_1), (a_2, b_2)) = d_X(a_1, a_2) + d_Y(b_1, b_2)$). The continuity of γ_X and γ_Y amount to the following: for each $t \in [0, 1]$ and $\varepsilon > 0$, there exists $\delta_1 > 0$ such that

$$|x - t| < \delta_1 \implies d_X(\gamma_X(x), \gamma_X(t)) < \frac{\varepsilon}{2},$$

and there exists $\delta_2 > 0$ such that

$$|y - t| < \delta_2 \implies d_Y(\gamma_Y(y), \gamma_Y(t)) < \frac{\varepsilon}{2}.$$

Then, if $\delta = \min(\delta_1, \delta_2)$, we find that $|s - t| < \delta$ implies that

$$\begin{aligned} d_{X \times Y}(\gamma(s), \gamma(t)) &= d_{X \times Y}((\gamma_X(s), \gamma_Y(s)), (\gamma_X(t), \gamma_Y(t))) \\ &= d_X(\gamma_X(s), \gamma_X(t)) + d_Y(\gamma_Y(s), \gamma_Y(t)) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

hence γ is continuous at every $t \in [0, 1]$, so it is continuous. This proves that X and Y being path connected implies that $X \times Y$ is path connected.

Now we show that if X and Y are connected, then so is $X \times Y$.

Lemma 10.1. Let U be a subset of $X \times Y$. For a given $y \in Y$, define $\pi_X(U, y)$ to be the set $\{x : (x, y) \in U\} \subseteq X$, and for a given $x \in X$, define $\pi_Y(U, x)$ to be the set $\{y : (x, y) \in U\} \subseteq Y$. Then, if U is open in $X \times Y$, then, for any $x \in X$ and $y \in Y$, $\pi_X(U, y)$ and $\pi_Y(U, x)$ are open in X and Y respectively.

Proof. Suppose U is open in $X \times Y$. Then, for some $y \in Y$, let $L_y = \{(x, y) : x \in X\}$. Observe that the induced metric on L_y is

$$d_{L_y}((a, y), (b, y)) = d_{X \times Y}((a, y), (b, y)) = d_X(a, b) + d_Y(y, y) = d_X(a, b).$$

We see that $L_y \cap U$ is the set

$$\{(x, y) : x \in \pi_X(U, y)\} = \pi_X(U, y) \times \{y\},$$

which is open in L_y with the induced metric. In particular, for any $a \in \pi_X(U, y)$, there exists an ε such that $B_{L_y}((a, y), \varepsilon) \subseteq L_y \cap U$. That is,

$$d_{L_y}((a, y), (x, y)) = d_X(a, x) < \varepsilon \implies (x, y) \in L_y \cap U = \pi_X(U, y) \times \{y\},$$

but this is the same thing as saying that $x \in \pi_X(U, y)$. To be clear, we just showed that, for each $a \in \pi_X(U, y)$, there exists an ε such that $d_X(a, x) < \varepsilon$ implies $x \in \pi_X(U, y)$, so $B_X(a, \varepsilon) \subseteq \pi_X(U, y)$. Thus, $\pi_X(U, y)$ is open in X .

The proof that $\pi_Y(U, x)$ is open in Y for any $x \in X$ is identical, so we are done. $\square$

This lemma is our mechanism for partitioning X and Y into open sets. Specifically, suppose $X \times Y$ is disconnected, so that it can be written as a union of nonempty disjoint sets A and B that are open in $X \times Y$; we want to show that one of X , Y must be disconnected. The lemma says that for any $x \in X$ and $y \in Y$, the four sets

$$\pi_X(A, y), \quad \pi_X(B, y), \quad \pi_X(A, x), \quad \pi_X(B, x)$$

are all open. Note also that $\pi_X(A, y)$ is the complement of $\pi_X(B, y)$ in X because they are obviously disjoint (since A and B are disjoint) and

$$\pi_X(A, y) \cup \pi_X(B, y) = \{x : (x, y) \in A \cup B\} = \{x : (x, y) \in X \times Y\} = X.$$

Similarly, $\pi_Y(A, x)$ is the complement of $\pi_Y(B, x)$ in Y . In other words, π_X and π_Y naturally partition X and Y into disjoint open sets. Thus, we just need to show that there either exists a y such that $\pi_X(A, y)$ and its complement $\pi_X(B, y)$ are both nonempty, or there exists an x such that $\pi_Y(A, x)$ and its complement $\pi_Y(B, x)$ are both nonempty. Equivalently, we

want $\pi_X(A, y)$ to be both nonempty and not equal to X , or we want $\pi_Y(A, x)$ to be both nonempty and not equal to Y .

Now, if we could find a y that made $\pi_X(A, y)$ both nonempty and not equal to X (so X is disconnected), then we'd be done, so suppose no such y exists. We will then prove that Y is disconnected. Observe that

$$\bigcup_{y \in Y} \pi_X(A, y) \times \{y\} = \bigcup_{y \in Y} \{(x, y) : (x, y) \in A\} = \{(x, y) : (x, y) \in A\} = A,$$

so if $\pi_X(A, y) = \emptyset$ for all y then A would be empty, and if $\pi_X(A, y) = X$ for all y then A would equal $X \times Y$, both of which contradict our assumptions about A . Therefore there exist y_1 and y_2 in Y such that $\pi_X(A, y_1) = \emptyset$ and $\pi_X(A, y_2) = X$, so we know that $(x, y_1) \notin A$ and $(x, y_2) \in A$ for any $x \in X$. Hence, for any choice of $x \in X$,

$$\pi_Y(A, x) = \{y : (x, y) \in A\},$$

so $y_1 \notin \pi_Y(A, x)$ but $y_2 \in \pi_Y(A, x)$. In particular $\pi_Y(A, x)$ is nonempty and not equal to Y (thus having nonempty complement), which proves that Y is disconnected. $\square$

Problem 11. In ℓ^2 consider the infinite-dimensional unit sphere:

$$S^\infty = \left\{ \mathbf{x} \in \ell^2 : \sum_{i=1}^{\infty} x_i^2 = 1 \right\}.$$

Is $S^\infty \subseteq \ell^2$ a connected subspace of ℓ^2 ?

Solution. We show that S^∞ is path connected. Let $\mathbf{x}$ and $\mathbf{y}$ be points in S^∞ such that $\mathbf{x} + \mathbf{y} \neq \mathbf{0}$. Then, I claim that the following function $\gamma: [0, 1] \rightarrow S^\infty$ is a path between $\mathbf{x}$ and $\mathbf{y}$:

$$\gamma(t) = \frac{(1-t)\mathbf{x} + t\mathbf{y}}{\|(1-t)\mathbf{x} + t\mathbf{y}\|}.$$

(The norm $\|\mathbf{z}\|$ is $d(\mathbf{z}, \mathbf{0}) = \sqrt{\sum_{n=1}^{\infty} z_n^2}$ and has roughly the same properties as the $\mathbb{R}^n$ norm, namely $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ and $\|c\mathbf{x}\| = |c| \cdot \|\mathbf{x}\|$ for any $c \in \mathbb{R}$.) Some preliminary checks:

- First,

$$\gamma(0) = \frac{\mathbf{x}}{\|\mathbf{x}\|} = \mathbf{x}, \quad \gamma(1) = \frac{\mathbf{y}}{\|\mathbf{y}\|} = \mathbf{y}$$

because the norms of $\mathbf{x}$ and $\mathbf{y}$ are both 1.

- Second, $\gamma(t)$ is always defined. The only concern is that the denominator may be zero, in which case $(1-t)\mathbf{x} + t\mathbf{y}$ would be the zero vector (denoted $\mathbf{0}$). If $t = 0$, then $\gamma(t) = \mathbf{x} \neq \mathbf{0}$, and, for $t \neq 0$,

$$(1-t)\mathbf{x} + t\mathbf{y} = \mathbf{0} \iff \mathbf{x} = \frac{1-t}{t}\mathbf{y}.$$

We know that $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$, so in order for the norms of $\mathbf{x}$ and $\frac{1-t}{t}\mathbf{y}$ to be equal, we must have $\frac{1-t}{t} = 1$, so $t = \frac{1}{2}$, which means that the denominator can only be zero if $\mathbf{x} + \mathbf{y} = \mathbf{0}$, and this is precisely the case I excluded.

- Third, $\gamma(t)$ is always in S^∞ because the norm of $\gamma(t)$ is just

$$\frac{1}{\|(1-t)\mathbf{x} + t\mathbf{y}\|} \cdot \|(1-t)\mathbf{x} + t\mathbf{y}\| = 1.$$

To prove that γ is indeed a path between $\mathbf{x}$ and $\mathbf{y}$, we have to show that γ is continuous. We have shown in Lemma L2 that $(1-t)\mathbf{x} + t\mathbf{y}$ is a continuous function of t . In fact the function $\mathbf{z} \mapsto \|\mathbf{z}\|$ is also continuous because, for each $\varepsilon > 0$, we can take $\delta = \varepsilon$ to get

$$d(\mathbf{x}, \mathbf{y}) < \varepsilon \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| = d(\mathbf{x}, \mathbf{y}) < \varepsilon,$$

where the inequality of norms follows from the triangle inequality: if WLOG $\|\mathbf{x}\| \geq \|\mathbf{y}\|$, then

$$\|\mathbf{x}\| = d(\mathbf{x}, \mathbf{0}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{0}) = \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|.$$

We can then show that γ is continuous as follows:

- First, $\|(1-t)\mathbf{x} + t\mathbf{y}\|$ is the composed map $[0, 1] \rightarrow \ell^2 \rightarrow \mathbb{R}$ given by

$$t \mapsto (1-t)\mathbf{x} + t\mathbf{y} \mapsto \|(1-t)\mathbf{x} + t\mathbf{y}\|,$$

which is continuous because the intermediate maps are both continuous, as per the result of Problem 3.

- Then, since $\|(1-t)\mathbf{x} + t\mathbf{y}\|$ is never zero for $t \in [0, 1]$, the quotient of continuous functions

$$\frac{(1-t)\mathbf{x} + t\mathbf{y}}{\|(1-t)\mathbf{x} + t\mathbf{y}\|} = \gamma(t)$$

is also a continuous function of t .

Therefore, we have shown that if $\mathbf{x}$ and $\mathbf{y}$ are not antipodal points on S^∞ , then they are connected by a path.

To finish, in the case where $\mathbf{x} = -\mathbf{y}$, pick an arbitrary third point $\mathbf{p}$ that is not equal to $\mathbf{x}$ or $\mathbf{y}$. Then $\mathbf{p}$ is not equal to $-\mathbf{x} = \mathbf{y}$ or $-\mathbf{y} = \mathbf{x}$, so $\mathbf{x}$ and $\mathbf{y}$ are both connected by a path to $\mathbf{p}$. By Lemma L3, this implies $\mathbf{x}$ and $\mathbf{y}$ are connected by a path as the relation of “connected by a path” is transitive.

Therefore, any two points in S^∞ are connected by a path, so S^∞ is path connected and thus connected. $\square$

Problem 12. Explain why $\mathbb{Q}$ is dense in $\mathbb{R}$ with the standard metric.

Solution. We have shown that any real number is the limit of some sequence of rational numbers, so every real number is a limit point of $\mathbb{Q}$; thus, $\overline{\mathbb{Q}} = \mathbb{R}$. $\square$

Problem 13. Show that

$$D = \{\mathbf{x} \in \mathbb{R}^\infty : x_i = 0 \text{ for all but finitely many } i \in \mathbb{N}\}$$

is dense in ℓ^1 .

Solution. Let $\mathbf{x}$ be in ℓ^1 and define $s = \sum_{k=1}^\infty |x_k|$. I will show that $\mathbf{x}$ is a limit point of D by proving that it is the limit of the sequence $(\mathbf{x}_n)$, where $\mathbf{x}_n$ is the sequence whose k th term is x_k when $k \leq n$, and 0 when $k > n$ (for example $\mathbf{x}_3 = (x_1, x_2, x_3, 0, 0, \dots)$). We see that

$$d(\mathbf{x}, \mathbf{x}_n) = \sum_{k=1}^n |x_k - x_k| + \sum_{k=n+1}^\infty |x_k - 0| = \sum_{k=n+1}^\infty |x_k| = s - \sum_{k=1}^n |x_k| = s - s_n,$$

where we define s_n to be the partial sums of the sequence $(|x_k|)$. Taking the limit, we see that

$$\lim_{n \rightarrow \infty} d(\mathbf{x}, \mathbf{x}_n) = \lim_{n \rightarrow \infty} s - s_n = s - \lim_{n \rightarrow \infty} s_n = s - s = 0,$$

so $d(\mathbf{x}, \mathbf{x}_n)$ will eventually be smaller than ε for any choice of $\varepsilon > 0$; thus, $(\mathbf{x}_n)$ converges to $\mathbf{x}$ in the metric space (ℓ^1, d) . $\square$

Problem 14. In the last homework we proved that ℓ^1 is a subset of ℓ^2 . Is ℓ^1 dense in ℓ^2 ?

Solution. In fact

$$D = \{\mathbf{x} \in \mathbb{R}^\infty : x_i = 0 \text{ for all but finitely many } i \in \mathbb{N}\} \subseteq \ell^1$$

is dense in ℓ^2 , too — this is enough to prove ℓ^1 is dense in ℓ^2 because a limit point of D is also a limit point of ℓ^1 .

What follows is the exact same proof as in the previous problem, but with ℓ^2 instead of ℓ^1 . Let $\mathbf{x}$ be in ℓ^2 and define $s = \sum_{k=1}^{\infty} x_k^2$. I will show that $\mathbf{x}$ is a limit point of D by proving that it is the limit of the sequence $(\mathbf{x}_n)$, where $\mathbf{x}_n$ is the sequence whose k th term is x_k when $k \leq n$, and 0 when $k > n$ (for example $\mathbf{x}_3 = (x_1, x_2, x_3, 0, 0, \dots)$). We see that

$$d(\mathbf{x}, \mathbf{x}_n) = \sum_{k=1}^n (x_k - x_k)^2 + \sum_{k=n+1}^{\infty} (x_k - 0)^2 = \sum_{k=n+1}^{\infty} x_k^2 = s - \sum_{k=1}^n x_k^2 = s - s_n,$$

where we define s_n to be the partial sums of the sequence (x_k^2) . Taking the limit, we see that

$$\lim_{n \rightarrow \infty} d(\mathbf{x}, \mathbf{x}_n) = \lim_{n \rightarrow \infty} s - s_n = s - \lim_{n \rightarrow \infty} s_n = s - s = 0,$$

so $d(\mathbf{x}, \mathbf{x}_n)$ will eventually be smaller than ε for any choice of $\varepsilon > 0$; thus, $(\mathbf{x}_n)$ converges to $\mathbf{x}$ in the metric space (ℓ^2, d) . $\square$

Problem 15. Let (X, d) be a metric space and $D \subseteq X$ a dense set. Suppose that $f: X \rightarrow \mathbb{R}$ is zero at all points of D . Show that if f is continuous then f is zero at all points of X .

Solution. Since D is a dense subset of X , every point of X is a limit point of D . This means that for each $x \in X$, there is a sequence (d_n) of elements of D that converges to x . By continuity, we thus have

$$f(x) = f\left(\lim_{n \rightarrow \infty} d_n\right) = \lim_{n \rightarrow \infty} f(d_n) = \lim_{n \rightarrow \infty} 0 = 0,$$

so $f(x) = 0$ for all $x \in X$. $\square$